

Chi-Square Tests & Inference for Regression



Chi-square goodness-of-fit test

- 1 A die is rolled 60 times, expecting each face to appear equally often (expected count 10 per face). The observed counts are 12, 8, 11, 9, 7, 13. Compute the chi-square test statistic using $\chi^2 = \sum \frac{(O - E)^2}{E}$.
 - 2 State the null and alternative hypotheses for the goodness-of-fit test in the die-rolling scenario above.
-

Chi-square test for independence

- 3 A two-way table examines whether preferred learning style (visual, auditory, kinesthetic) is associated with academic major (STEM, non-STEM). State the null and alternative hypotheses for a chi-square test of independence in this context.
 - 4 In a chi-square test of independence with a 3×2 table (3 learning styles, 2 major categories), find the degrees of freedom, using $df = (\text{rows} - 1)(\text{columns} - 1)$.
-

Interpreting chi-square results

- 5 A chi-square test of independence produces a test statistic $\chi^2 = 8.2$ with $df = 2$, and a p -value of 0.017. State a conclusion at the $\alpha = 0.05$ significance level, in context.
 - 6 Explain a condition that must be checked before conducting a chi-square test, related to expected cell counts.
-

Inference for the slope of a regression line

- 7 A regression of exam score on hours studied gives a sample slope $b_1 = 6.2$ with standard error $SE_{b_1} = 1.1$, based on $n = 25$ data points. Construct a 95% confidence interval for the true population slope β_1 , using a critical value $t^* \approx 2.069$ (with $df = 23$).
 - 8 State the null and alternative hypotheses for testing whether hours studied has a significant linear relationship with exam score, and compute the test statistic using the values from the previous problem.
-

Conditions for regression inference

- 9 State the four conditions (sometimes remembered as LINER: Linear, Independent, Normal, Equal variance, Random) required for valid inference about a regression slope.
-

Putting it together: a full inference scenario

- 10 A study finds a chi-square test statistic of $\chi^2 = 3.1$ with $df = 1$, and a critical value of $\chi^2_{0.05} = 3.84$ at the $\alpha = 0.05$ level. State the conclusion, and explain how this differs from a p -value based approach.
-

Chi-square test for homogeneity

- 11 Explain the difference between a chi-square test of independence and a chi-square test of homogeneity, even though both use the same formula and table structure.
-

Computing expected counts for a chi-square test

- 12 In a two-way table, a row total is 80, a column total is 120, and the grand total is 300. Find the expected count for that cell, using $E = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$.
- 13 State the degrees of freedom for a chi-square goodness-of-fit test with 6 categories, using $df = (\text{number of categories}) - 1$.
- 14 A genetics experiment expects offspring to appear in a 9:3:3:1 ratio across four phenotype categories, based on 160 total offspring observed. Find the expected count for each category, and briefly describe what the chi-square goodness-of-fit test would examine using these expected counts.
- 15 A study examines the relationship between hours of sleep per night (x) and reaction time in milliseconds (y) for 30 participants, finding a regression slope $b_1 = -8.5$ with standard error $SE_{b_1} = 2.3$. State the hypotheses for testing whether more sleep is associated with a significant decrease in reaction time (a one-sided test), compute the test statistic, and interpret the sign of the slope in context.
- 16 A university surveys 240 students, recording both their preferred study location (library, dorm, coffee shop) and their class year (underclassman, upperclassman), to test whether study location preference is independent of class year. The resulting chi-square test statistic is $\chi^2 = 11.4$ with $df = 2$, and the corresponding p -value is 0.0033. State the conclusion at $\alpha = 0.01$, and explain, in the context of this scenario, what rejecting or failing to reject H_0 would mean for the university's understanding of its students.

- 17 A researcher studies the relationship between advertising spending (in thousands of dollars, x) and monthly sales (in thousands of dollars, y) for 22 small businesses, obtaining a regression slope $b_1 = 3.2$ with standard error $SE_{b_1} = 0.85$. Construct a 95% confidence interval for the true population slope β_1 using a critical value $t^* \approx 2.086$ (with $df = 20$), and interpret the interval in context, including whether it suggests advertising spending has a statistically significant positive association with sales.